

Class 6 — Set B

Mateng EduFest 2026

1. Which of the following statements is always true?

- A) Every composite number is even.
- B) Every prime number is odd.
- C) Every whole number except 0 has at least one factor other than 1.
- D) Every prime number has exactly two distinct factors.

2. Which of the following statements about factors and multiples is correct?

- A) Every factor of a number is greater than the number.
- B) Every multiple of a number is smaller than the number.
- C) A number can have infinitely many multiples but only finitely many factors.
- D) A number has infinitely many factors.

3. Which of the following fractions lies between $\frac{3}{5}$ and $\frac{4}{5}$?

- A) $\frac{5}{9}$
- B) $\frac{7}{10}$
- C) $\frac{1}{2}$
- D) $\frac{9}{10}$

4. Which of the following decimals is nearest to 1?

- A) 0.89
- B) 0.909
- C) 0.99
- D) 0.999

5. The average of six consecutive natural numbers is 18.5. The largest number is:

- A) 20
- B) 21
- C) 22
- D) 23

6. The average of five numbers is 24. If one number is excluded, the average of the remaining four numbers becomes 20. The excluded number is:

- A) 36
- B) 38
- C) 40
- D) 44

7. The ratio of boys to girls in a class is 5:4. Which of the following cannot represent the number of students in the class?

- A) 18
- B) 27
- C) 36
- D) 40

8. If $x : y = 3 : 5$ and $y : z = 10 : 7$, then $x : y : z$ is:

- A) 3 : 10 : 7
- B) 6 : 10 : 7
- C) 3 : 5 : 7
- D) 6 : 5 : 7

9. The simple interest on a sum depends on:

- A) Principal and Time only
- B) Principal and Rate only
- C) Rate and Time only
- D) Principal, Rate and Time

10. Which of the following statements is false?

- A) Every prime number greater than 2 is odd.
- B) Every even number greater than 2 is composite.
- C) The number 1 is a prime number.
- D) Every multiple of a number is divisible by that number.

11. Which of the following statements is always true for prime numbers?

- A) They are divisible only by themselves.
- B) They have exactly two distinct factors.
- C) They are always odd.
- D) They are not multiples of themselves.

12. Which of the following statements regarding even numbers is correct?

- A) Every even number is composite.
- B) Every even number greater than 2 is composite.
- C) Every even number is a multiple of 4.
- D) Every even number is prime.

13. Which of the following statements about fractions is correct?

- A) Every proper fraction is greater than 1.
- B) Every improper fraction is less than 1.
- C) In a proper fraction, the numerator is less than the denominator.
- D) In an improper fraction, the numerator is always equal to the denominator.

14. Which of the following statements is true about equivalent fractions?

- A) They have different values.
- B) They represent the same part of a whole.
- C) Their denominators are always equal.
- D) Their numerators are always equal.

15. Which of the following pairs is correctly matched?

- A) HCF of fractions = $\text{HCF}(\text{numerators}) / \text{HCF}(\text{denominators})$
- B) LCM of fractions = $\text{LCM}(\text{numerators}) / \text{LCM}(\text{denominators})$
- C) HCF of fractions = $\text{HCF}(\text{numerators}) / \text{LCM}(\text{denominators})$
- D) LCM of fractions = $\text{HCF}(\text{numerators}) / \text{LCM}(\text{denominators})$

16. Which of the following formulas is incorrect for finding the Principal in simple interest?

- A) $P = A - SI$
- B) $P = SI \times 100 / (r \times t)$
- C) $P = A \times 100 / (100 + r \times t)$
- D) None of the above

17. Which of the following formulas is correct?

- A) $SP = CP \times (100 + L\%) / 100$
- B) $SP = CP \times (100 - P\%) / 100$
- C) $CP = SP \times 100 / (100 + L\%)$
- D) None of the above

18. If a number is divisible by 72, then it must also be divisible by:

- A) 6 and 8 only
- B) 8 and 9 only
- C) 3, 8 and 9 only
- D) 2, 3, 4, 6, 8 and 9

19. The first term of a ratio is called:

- A) Consequent
- B) Quotient
- C) Antecedent
- D) Dividend

20. Which of the following statements is false?

- A) Every multiple of a number is divisible by that number.
- B) Every factor of a number divides the number exactly.
- C) Every prime number is a composite number.
- D) Zero is an even number.

21. Simplify: $4 \text{ of } [32 \div 8 \{ 25 - (19 + 64 \div 16 \times 2) \}]$

- A) 2
- B) 4
- C) 8
- D) 12

22. Simplify: $8 \frac{1}{2} - [3 \frac{1}{5} - 4 \frac{1}{2} \text{ of } 5 \frac{1}{3} + \{ 11 - (3 - 1 \frac{1}{4} - 5/8) \}]$

- A) $-3 \frac{11}{20}$
- B) $-120/31$
- C) $120/31$
- D) $3 \frac{11}{20}$

23. A store sells a packet of 5 apples for ₹25 and a single apple for ₹6. If a lady purchases 27 apples, how much money will she pay?

- A) ₹128
- B) ₹130
- C) ₹137
- D) ₹150

24. Rajesh's weight is 5 kg less than Ram's weight, and Neha's weight is 3 kg more than Ram's weight. If the total weight of the three is 103 kg, then the weight of Ram is

- A) 34 kg
- B) 38 kg
- C) 33 kg
- D) 35 kg

25. The unit digit in the product of $163 \times 87^* \times 239$ is 1. The digit at the place of * will be

- A) 1
- B) 3
- C) 7
- D) 9

26. The number 7254^*38 is divisible by 9. The digit at the place of * is

- A) 4
- B) 7
- C) 6
- D) 5

27. There are 500 eggs in a box. $\frac{3}{25}$ of them got broken, and $\frac{4}{5}$ of the remaining eggs were sold. The number of eggs left is

- A) 80
- B) 88
- C) 40
- D) 36

28. In a class of 30 students, the number of girls is $\frac{1}{5}$ th of the number of boys. How many boys are there in the class?

- A) 25
- B) 18
- C) 20
- D) 19

29. In what time will the simple interest on ₹4000 at 10% per annum be the same as the simple interest on ₹10000 for 4 yr at 4% per annum?

- A) 3 yr
- B) 4 yr
- C) 5 yr
- D) 6 yr

30. Which of the following lists the numbers 1219, 1225, 1229 and 1237 in ascending order?

- A) 1219, 1237, 1229, 1225
- B) 1237, 1229, 1219, 1225
- C) 1237, 1229, 1225, 1219
- D) 1219, 1225, 1229, 1237

31. By how much does $6\frac{7}{8}$ exceed $6\frac{7}{8}$?

- A) $6\frac{2}{3}$
- B) $6\frac{3}{4}$
- C) $7\frac{1}{2}$
- D) $8\frac{3}{4}$

32. What is the prime factorization of 37800?

- A) $2 \times 2 \times 3 \times 3 \times 5 \times 5 \times 7 \times 7$
- B) $2 \times 2 \times 2 \times 3 \times 3 \times 3 \times 5 \times 5 \times 7$
- C) $8 \times 27 \times 25 \times 7$
- D) $2 \times 4 \times 25 \times 27 \times 7$

33. Raju deposited ₹23.25 in the first week, ₹27.50 in the second week and ₹30.75 in the third week. He had deposited ₹250.60 earlier. What is his total amount in the bank now?

- A) ₹350
- B) ₹332.10
- C) ₹325.75
- D) ₹275.25

34. If $\sqrt{136.18} = 0.0276$, then the value of $\sqrt{(1/0.0003618)}$ will be

- A) 276
- B) 2760
- C) 27600
- D) 0.0002760

35. 12 men or 15 women can do a piece of work in 21 days. Find the number of days required to complete the same work with 6 men and 10 women.

- A) 15
- B) 18
- C) 21
- D) 24

36. In a family of 10 persons, the father, mother and eldest son earn ₹4000, ₹3000 and ₹4400 per month respectively. What is the average monthly income per family member?

- A) ₹1260
- B) ₹1600
- C) ₹1140
- D) ₹3800

37. The average temperature from Monday to Thursday was 48° . The average temperature from Tuesday to Friday was 52° . If the temperature on Monday was 42° , what was the temperature on Friday?

- A) 56°
- B) 54°
- C) 58°
- D) 60°

38. The average age of 5 officers in a department is 32 yr. If the age of their supervisor is added, the average increases by 1 yr. What is the supervisor's age?

- A) 32 yr
- B) 48 yr
- C) 38 yr
- D) 42 yr

39. A student scored 18 out of 25 marks in the first test of Maths, and 22 out of 25 marks in the second test. By how many percentage points does his second-test score exceed his first-test score?

- A) 4%
- B) 8%
- C) 16%
- D) 80%

40. Out of a total of 250 marks, a student got 30% marks and failed by 25 marks. The marks necessary for passing are

- A) 50
- B) 75
- C) 100
- D) 125

41. An article is sold for ₹500, and a loss is incurred. Had the article been sold for ₹700, the shopkeeper would have gained three times the former loss. What is the cost price of the article?

- A) ₹525
- B) ₹550
- C) ₹600
- D) ₹650

42. A shopkeeper bought 15 tables at ₹500 each and 20 chairs at ₹300 each, and spent ₹40 on transportation. He sold the tables and chairs at a flat rate of ₹380 each. What is his gain or loss?

- A) ₹240, loss
- B) ₹240, gain
- C) ₹250, loss
- D) ₹250, gain

43. If $x : y = 7 : 5$, find the value of $(5x - 2y) : (3x + 2y)$.

- A) 24 : 13
- B) 25 : 31
- C) 23 : 25
- D) 26 : 31

44. If an article is sold at a loss of 50%, find the cost price in terms of the selling price.

- A) $\frac{1}{2}$
- B) 2
- C) 2.5
- D) None of these

45. At 25% per annum simple interest, in how many years will the amount become three times the principal?

- A) 4
- B) 6
- C) 8
- D) 10

Questions 46–50

Questions 46–50: There are two or more correct options.

46. Which of the following statements about average are correct?

- A) Average is obtained by dividing the sum of observations by the number of observations.
- B) The average need not be one of the observations.
- C) The average is always greater than the largest observation.
- D) The average can be a decimal number.

47. Which of the following statements about ratio and proportion are correct?

- A) The first term of a ratio is called the antecedent.
- B) Equal ratios form a proportion.
- C) In a proportion, the product of extremes equals the product of means.
- D) Every pair of ratios forms a proportion.

48. A shopkeeper buys an article for ₹500 and sells it for ₹600. Which of the following statements are correct?

- A) The shopkeeper makes a profit.
- B) The profit is ₹100.
- C) The Cost Price is ₹600.
- D) The Selling Price is greater than the Cost Price.

49. A sum of ₹1000 is invested at simple interest for 2 years at 5% per annum. Which of the following statements are correct?

- A) The principal is ₹1000.
- B) The interest earned in one year is ₹50.
- C) The total simple interest for 2 years is ₹100.
- D) The amount after 2 years is ₹1050.

50. In a class, the ratio of boys to girls is 3:2 and there are 15 boys. Which of the following statements are correct?

- A) The number of girls is 10.
- B) The total number of students is 25.
- C) The ratio of girls to boys is 2:3.
- D) The number of girls is 12.